

MODULE 3 | PART 1

GEOMETRY

IOQM & RMO 2026

GEOMETRY

Euclid's Definitions

- ❖ A point is that which has no part.
- ❖ A line is breadthless length.
- ❖ The ends of a line segment are points.
- ❖ A straight line is that which has length only.
- ❖ A surface is that which has length and breadth only.
- ❖ The edges of a surface are lines.
- ❖ A plane surface is that which lies evenly with the straight lines on itself.

Some Geometrical Concepts

Axioms: The basic facts that are taken for granted, without proof, are called axioms.

E.g. Halves of equals are equal.

Statements: A sentence that can be judged to be true or false is called a statement.

E.g. The sum of the angles of a triangle is 180° , is a true statement.

Theorems: A statement that requires a proof is called a theorem. Establishing the truth of a theorem is known as proving the theorem.

E.g. The sum of all angles around a point is 360° .

Corollary: A statement whose truth can easily be decided from a theorem is called its corollary.

Conjecture: In mathematics, a conjecture is a conclusion or proposition based on incomplete information, for which no proof has been found.

E.g. Goldbach Conjecture states that every even integer greater than two is the sum of two prime numbers.

Some Terms Related to Geometry

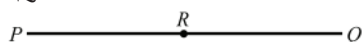
Point: A point is an exact location. A fine dot represents a point. Point is denoted by capital letter A, B, P, Q, R etc.

Line segment: The straight path between two points A and B is called the line segment $\overline{AB}$. The points A and B are called the end points of the line segment $\overline{AB}$. A line segment has a definite length.

Line: A line segment $\overline{AB}$ when extended indefinitely in both direction is called the line $\overleftrightarrow{AB}$. A line has no end points. A line has no definite length. Sometimes, we label lines by small letters l, m, n etc.



Interior point of a line segment: A point R is called an interior point of a line segment $\overline{PQ}$ if R lies between P and Q but R is neither P nor Q .



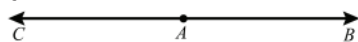
Congruence of line segments: Two line segments $\overline{AB}$ and $\overline{CD}$ are congruent if trace copy of one can be superposed on the other so as to cover it completely and exactly. In this case we write $\overline{AB} \cong \overline{CD}$. In other words, we can say two lines are congruent if their lengths are same.

Distance between two points: The distance between two points P and Q is the length of line segment $\overline{PQ}$.

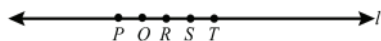
Ray: Directed line segment is called a ray. If $\overrightarrow{AB}$ is a ray then, point A is called initial point of the ray.



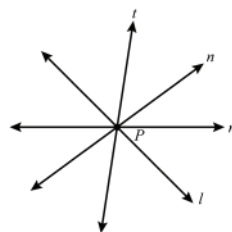
Opposite rays: Two rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are said to be opposite rays if they are collinear and point A is the only common point of the two rays.



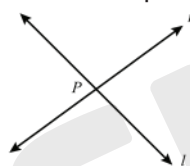
Collinear points: Three or more points are said to be collinear if there is a line which contains all of them.



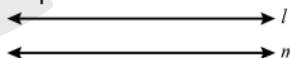
Concurrent Lines: Three or more lines are said to be concurrent if there is a point which lies on all of them.



Intersecting lines: Two lines are intersecting if they have a common point. The common point is called the 'point of intersection'.



Parallel lines: Two lines l and m in a plane are said to be parallel lines if they do not have a common point.



Plane: A plane is a surface such that every point of the line joining any two points on it lies on it.

Euclid's Axioms

- ❖ Things which are equal to the same thing are equal to one another.
- ❖ If equal are added to equals, the wholes are equal.
- ❖ If equal are subtracted from equals, the remainders are equal.
- ❖ The things that coincide with one another are equal to one another.
- ❖ The whole is greater than the part.

- ❖ The things that are double of the same thing are equal to one another.
- ❖ The things that are halves of the same thing are equal to one another.

Euclid's Five Postulates

- ❖ A straight line may be drawn from any one point to any other point.
- ❖ A terminated line or a line segment can be produced infinitely.

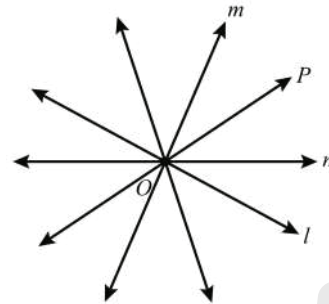


- ❖ A circle can be drawn with any centre and of any radius.
- ❖ All right angles are equal to one another.
- ❖ If a straight line falling on two straight lines makes interior angles on the same side of it taken together less than two right angles, then the two straight lines if produced infinitely will meet on that side on which the sum of angles are less than two right angles.



Incidence Axioms

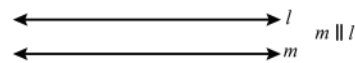
- ❖ A line is the collection of infinite number of points.
- ❖ Through a given point, an infinite number of lines can be drawn.



- ❖ Given two distinct points, there is one and only one line that contains both the points.

Parallel Axioms

If P is a point outside a line ℓ , then one and only one line can be drawn through P that is parallel to ℓ .

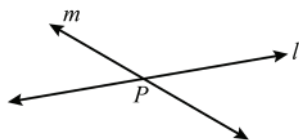


The above axioms may also be re-stated as follows: "Two intersecting lines cannot both be parallel to the same line".



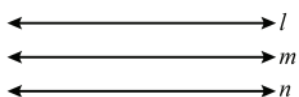
Some more results

- ❖ Two distinct lines cannot have more than one point in common.



- ❖ Two lines that are both parallel to the same line, are parallel to each other.

$$\text{i.e., } \ell \parallel n, m \parallel n \Rightarrow \ell \parallel m$$



1. If lines AB, AC, AD , and AE are parallel to a line ℓ , then prove that points A, B, C, D , and E are collinear.



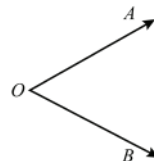
Lines and Angles

Lines

A line has length but no width and no thickness.

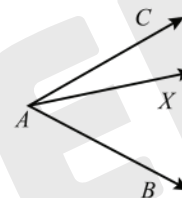
Angles

An angle is the union of two rays with a common initial point. The common initial point is called the '**vertex**' of the angle and two rays are called the '**arms**' of the angles



Angle Addition Axiom: If X is a point in the interior of $\angle BAC$, then

$$\angle BAC = \angle BAX + \angle XAC.$$



Note:

Every angle has a measure and a unit of measurement, which is typically degrees

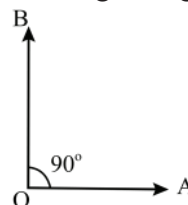
$$\text{One right angle} = 90^\circ$$

$$1^\circ = 60' \text{ (minutes)}$$

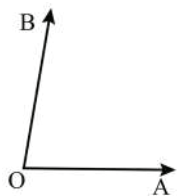
$$1' = 60'' \text{ (seconds)}$$

Types of Angles

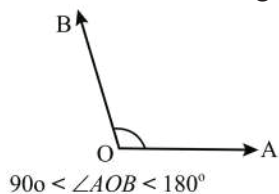
- ❖ **Right Angle:** An angle whose measure is 90° is called a right angle.



- ❖ **Acute Angle:** An angle whose measure is less than 90° is called an acute angle.



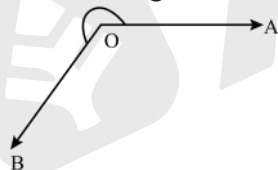
- ❖ **Obtuse Angle:** An angle whose measure is more than 90° , but less than 180° , is called an obtuse angle.



- ❖ **Straight Angle:** An angle whose measure is 180° is called a straight angle.

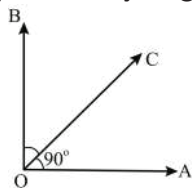


- ❖ **Reflex Angle:** An angle whose measure is more than 180° , but less than 360° is called a reflex angle.



$$180^\circ < \angle AOB < 360^\circ.$$

- ❖ **Complementary Angles:** Two angles, the sum of whose measures is 90° are called complementary angles.



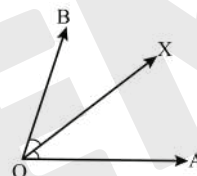
$\angle AOC$ and $\angle BOC$ are complementary as their sum is 90° .

- ❖ **Supplementary Angles:** Two angles, the sum of whose measures is 180° , are called the supplementary angles.



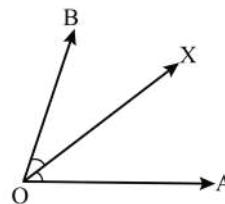
$\angle AOC$ and $\angle BOC$ are supplementary as their sum is 180° .

- ❖ **Angle Bisectors:** A ray OX is said to be the bisector of $\angle AOB$, if X is a point in the interior of $\angle AOB$, and $\angle AOX = \angle BOX$.



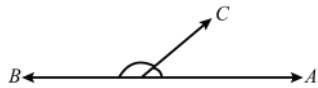
- ❖ **Adjacent Angles:** Two angles are called adjacent angles, if

- they have the same vertex,
- they have a common arm,
- non-common arms are on either side of the common arm.



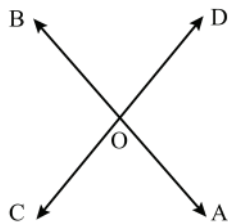
$\angle AOX$ and $\angle BOX$ are adjacent angles, OX is the common arm, OA and OB are the non-common arms and lie on either side of OX .

- ❖ **Linear Pair of Angles:** Two adjacent angles are said to form a linear pair of angles, if their non-common arms are two opposite rays.



$$\angle AOC + \angle BOC = 180^\circ.$$

- ❖ **Vertically Opposite Angles:** Two angles are called a pair of vertically opposite angles if their arms form two pairs of opposite rays.



$\angle AOC$ and $\angle BOD$ form a pair of vertically opposite angles. Also $\angle AOD$ and $\angle BOC$ form a pair of vertically opposite angles.

If two lines intersect, then the vertically opposite angles are equal, i.e.,

$$\angle AOC = \angle BOD \text{ and } \angle BOC = \angle AOD.$$

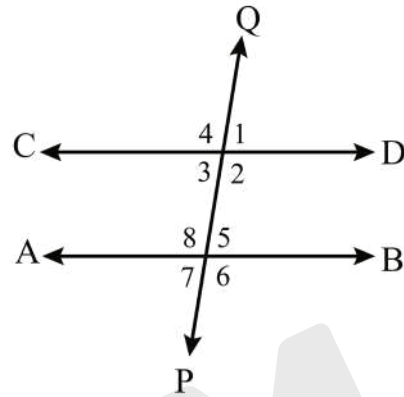


- Find the measure of an angle, if six times its complement is 12° less than twice its supplement.



Angles made by a Transversal

Transversal: A line that intersects two or more given lines at distinct points is called a transversal of the given lines.



- ❖ **Corresponding angles:** Two angles on the same side of a transversal are known as the corresponding angles if both lie either above the two lines or below the two lines. In figure $\angle 1$ and $\angle 5$, $\angle 4$ and $\angle 8$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$ are the pairs of corresponding angles.

If a transversal intersects two parallel lines then the corresponding angles are equal i.e.,

$$\angle 1 = \angle 5, \angle 4 = \angle 8, \angle 2 = \angle 6 \text{ and } \angle 3 = \angle 7.$$

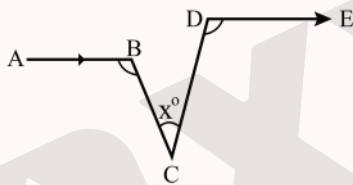
- ❖ **Alternate interior angles:** $\angle 3$ and $\angle 5$, $\angle 2$ and $\angle 8$, are the pairs of alternate interior angles.

If a transversal intersects two parallel lines then the each pair of alternate interior angles are equal i.e., $\angle 3 = \angle 5$ and $\angle 2 = \angle 8$.

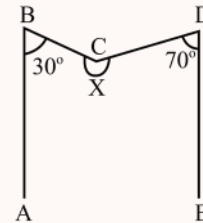
❖ **Co-interior angles:** The pair of interior angles on the same side of the transversal are called pairs of consecutive or co-interior angles. In figure $\angle 2$ and $\angle 5$, $\angle 3$ and $\angle 8$, are the pairs of co-interior angles. If a transversal intersects two parallel lines then each pair of consecutive interior angles are supplementary, i.e., $\angle 2 + \angle 5 = 180^\circ$ and $\angle 3 + \angle 8 = 180^\circ$.



3. In the adjoining figure, $\angle ABC = 100^\circ$, $\angle EDC = 120^\circ$, and $AB \parallel DE$. Then, find $\angle BCD$.



4. In figure, $AB \parallel ED$ and $\angle ABC = 30^\circ$, $\angle EDC = 70^\circ$ then find x° .



Triangle

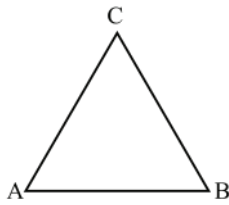
A plane figure bounded by three lines in a plane is called a triangle. Every triangle have three sides and three angles. If ABC is any triangle then AB, BC and CA are three sides and $\angle A, \angle B$ and $\angle C$ are three angles.

Types of Triangles

On the basis of sides, we have three types of triangles.

- ❖ **Scalene triangle:** A triangle in which no two sides are equal is called a scalene triangle.
- ❖ **Isosceles triangle:** A triangle having two sides equal is called an isosceles triangle.

- ❖ **Equilateral triangle:** A triangle in which all sides are equal is called an equilateral triangle.



On the basis of angles, we have three types:

- ❖ **Right triangle:** A triangle in which any one angle is right angle is called a right triangle.
- ❖ **Acute triangle:** A triangle in which all angles are acute is called an acute triangle.
- ❖ **Obtuse triangle:** A triangle in which any one angle is obtuse is called an obtuse triangle.

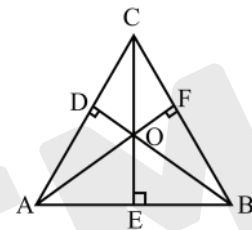


Some Important Terms

Altitudes

The perpendicular drawn from the opposite vertex to a side in a triangle is called an altitude of the triangle.

- ❖ There are three altitudes in a triangle.
- ❖ Point of intersection of the altitudes of a triangle is called the orthocentre.

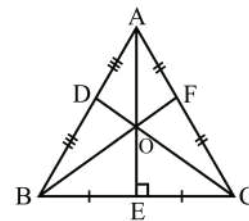


$AE, CD,$ and BF are the altitudes

- ❖ $\angle BOC = 180^\circ - \angle A$
 $\angle COA = 180^\circ - \angle B$
 $\angle AOB = 180^\circ - \angle C$

Medians

The line segment joining the mid-point of a side to the vertex opposite to the side is called a median.



$AE, CD,$ and BF are the medians
 $(BE = CE, AD = BD, AF = CF)$

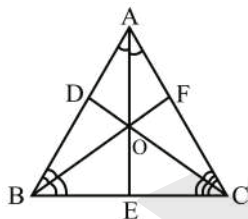
- ❖ There are three medians in a triangle.
- ❖ A median bisects the area of the triangle i.e.,

$$\text{Ar}(\triangle ABE) = \text{Ar}(\triangle AEC)$$

$$= \frac{1}{2} \text{Ar}(\triangle ABC)$$
- ❖ Point of intersection of medians is called centroid.

Angle Bisector

A line segment that originates from a vertex and bisects the same angle is called an angle bisector

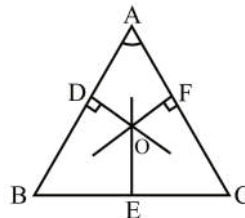


AE, CD , and BF are the angles' bisectors

- ❖ $\angle BAE = \angle CAE = \frac{1}{2} \angle BAC$ etc.
- ❖ Point of intersection of angle bisectors is called incentre.

Perpendicular Bisector

A line segment that bisects a side perpendicularly (i.e., at right angle) is called a perpendicular bisector of a side of triangle.



DO, EO , and FO are the perpendicular bisectors

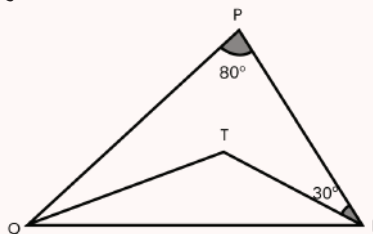
- ❖ All points on the perpendicular bisector of a line are equidistant from the ends of the line.
- ❖ Point of intersection of perpendicular bisectors is called circumcentre.

Euler Line

The circumcentre (S), the centroid (G) and orthocentre (H) of a nonequilateral triangle are collinear and $HG = 2GS$. The line passing through H, G, S is called the Euler line.



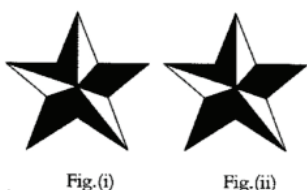
5. In figure, TQ and TR are the bisectors of $\angle Q$ and $\angle R$, respectively. If $\angle QPR = 80^\circ$ and $\angle PRT = 30^\circ$, determine $\angle TQR$ and $\angle QTR$.





Congruent Figures

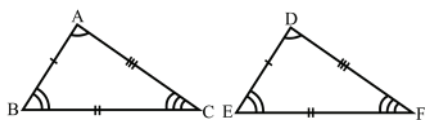
Two figures are called congruent if they have same shape and same size. In other words, two figures are called congruent if they have equal length, width and height.



In the above figures {fig.(i) and fig.(ii)}, both are equal in length, width, and height, so these are congruent figures.

Congruent Triangles

- ❖ Two triangles are congruent if and only if one of them can be made to superimpose on the other, so as to cover it exactly.



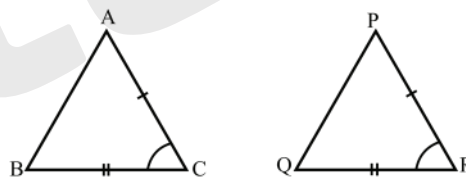
- ❖ If triangles $\triangle ABC$ and $\triangle DEF$ are congruent then $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$ and $AB = DE$, $BC = EF$, $AC = DF$.

- ❖ If two $\triangle ABC$ and $\triangle DEF$ are congruent then we write $\triangle ABC \cong \triangle DEF$, we cannot write as $\triangle ABC \cong \triangle DFE$ or $\triangle ABC \cong \triangle EDF$.

Hence, we say that two triangles are congruent if and only if there exists a one-one correspondence between their vertices such that the corresponding sides and the corresponding angles of the two triangles are equal.

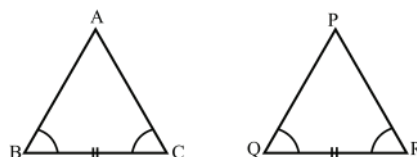
Criteria for Congruence of Two Triangles

- ❖ **SAS Congruence Criterion:**



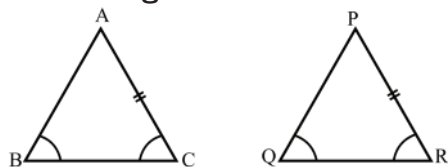
Two triangles are congruent if two sides and the included angle of one are equal to the corresponding sides and the included angle of the other triangle.

- ❖ **ASA Congruence Criterion:**



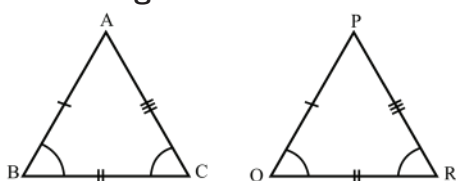
Two triangles are congruent if two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangle.

❖ **AAS Congruence Criterion:**



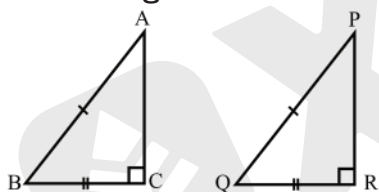
If any two angles and a non-included side of one triangle are equal to the corresponding angles and side of another triangle, then the two triangles are congruent.

❖ **SSS Congruence Criterion:**



Two triangles are congruent if the three sides of one triangle are equal to the corresponding three sides of the other triangle.

❖ **RHS Congruence Criterion:**



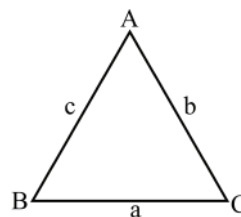
Two right triangles are congruent if the hypotenuse and one side of one triangle are, respectively, equal to the hypotenuse and one side of the other triangle.

Note:

If two triangles are congruent, then their corresponding sides and angles are also congruent by CPCT (corresponding parts of congruent triangles are also congruent).

Inequalities in a Triangle

In triangle ABC , $AB = c$, $BC = a$, and $CA = b$, then,



- ❖ The sum of any two sides of a triangle is greater than its third side,
i.e., $a + b > c$, $b + c > a$, $a + c > b$
- ❖ If two sides of a triangle are unequal, then the longer side has greater angle opposite to it.
i.e., if $AB > AC$ then $\angle C > \angle B$.
- ❖ Of all the line segments that can be drawn to a given line, from a point, not lying on it, the perpendicular line segment is the shortest
- ❖ The difference between any two sides of a triangle is less than its third side, i.e., in $\triangle ABC$,
 $a - b < c$, $b - c < a$, $a - c < b$
- ❖ In a right-angled triangle, the sum of squares of two smaller sides is equal to the square of its third side.
i.e., $a^2 + b^2 = c^2$ (where a and b are smaller compared to c).
- ❖ If sum of squares of two smaller sides is greater than the square of its third side, then that triangle is acute-angled triangle.
i.e., $a^2 + b^2 > c^2$ (where a and b are smaller compared to c).

❖ If sum of squares of two smaller sides is lesser than the square of its third side, then that triangle is obtuse-angled triangle.

i.e., $a^2 + b^2 < c^2$ (where a and b are smaller as compared to c).



6. Which of the following triplets make the sides of a triangle?

(i) $\{15, 7, 8\}$

(ii) $\{3.5, 4.5, 5.5\}$



7. AD, BE , and CF be the medians of a $\triangle ABC$. Prove that

$$2(AD + BE + CF) < 3(AB + BC + CA) < 4(AD + BE + CF).$$



8. In $\triangle PQR$, S is any point on the side QR . Show that $PQ + QR + RP > 2PS$.



Similar Triangles

Two triangles ABC and DEF are said to be similar if their

(i) Corresponding angles are equal.



$$\text{i.e., } \angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

(ii) Corresponding sides are proportional,

$$\text{i.e., } \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}.$$

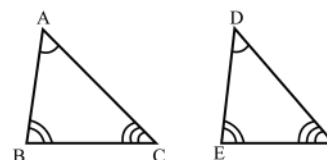
Criteria for Similar Triangles

AAA Similarity

If in two triangles, corresponding angles are equal, i.e., the two triangles are equiangular, then the triangles are similar.

$$\angle A = \angle D, \angle B = \angle E \text{ and } \angle C = \angle F$$

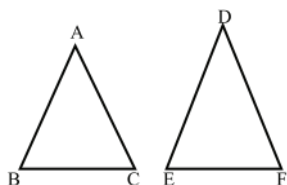
$$\Leftrightarrow \triangle ABC \sim \triangle DEF$$



SSS Similarity

If the corresponding sides of two triangles are proportional, then they are similar.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} \Leftrightarrow \triangle ABC \sim \triangle DEF$$

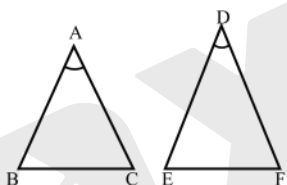


SAS Similarity

If in two triangles, one pair of corresponding sides are proportional and the included angles are equal, then the two triangles are similar.

$$\frac{AB}{DE} = \frac{AC}{DF} \text{ and } \angle BAC = \angle EDF$$

$$\Leftrightarrow \triangle ABC \sim \triangle DEF$$



Characteristic Properties of Similar Triangles

- ❖ If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding medians.
- ❖ If two triangles are equiangular, then the ratio of the corresponding sides is same as the ratio of the corresponding angle bisectors.
- ❖ If two triangles are equiangular, then the ratio of the corresponding sides is same as the ratio of the corresponding altitudes.
- ❖ If one angle of a triangle is equal to one angle of another triangle and the bisectors of these equal angles divide the opposite side in the same ratio, then the triangles are similar.
- ❖ If two sides and a median bisecting one of these sides of a triangle are, respectively, proportional to the two sides and the corresponding median of another triangle, then the triangles are similar.

Note:

If the ratio of sides of a triangle is $a : b : c$, then

ratio of their altitudes is $\frac{1}{a} : \frac{1}{b} : \frac{1}{c}$.



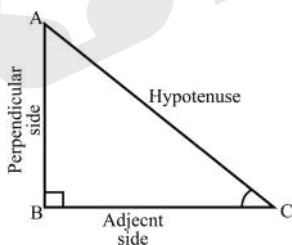
9. In quadrilateral $ABCD$,
 $\angle BAD \cong \angle ADC$ and $\angle ABD \cong \angle BCD$,
 $AB=8$, $BD=10$, and $BC=6$. The length CD
 may be written in the form $\frac{m}{n}$, where m and
 n are relatively prime positive integers. Find
 $m + n$.

[AIME]



Trigonometric Ratios

Consider a right-angled triangle, right-angled
 at B .



- ❖ **Sine:** Sine of an angle is defined as the ratio of the side opposite (perpendicular side) to that angle to the hypotenuse.
- ❖ **Cosine:** Cosine of an angle is defined as the ratio of the side adjacent to that angle to the hypotenuse.

- ❖ **Tangent:** Tangent of an angle is defined as the ratio of the side opposite to that angle to the side adjacent to that angle.
- ❖ **Cosecant:** Cosecant is the multiplicative inverse of sine.
- ❖ **Secant:** Secant is the multiplicative inverse of cosine.
- ❖ **Cotangent:** Cotangent is the multiplicative inverse of tangent.

With respect to $\angle C$, the ratios of trigonometry are given as:

- ❖ $\sin C = \frac{\text{Side opposite } \angle C}{\text{Hypotenuse}} = \frac{AB}{AC}$
- ❖ $\cos C = \frac{\text{Side adjacent } \angle C}{\text{Hypotenuse}} = \frac{BC}{AC}$
- ❖ $\tan C = \frac{\text{Side opposite to } \angle C}{\text{Side adjacent to } \angle C}$
 $= \frac{AB}{BC} = \frac{\sin \angle C}{\cos \angle C}$
- ❖ $\operatorname{cosec} C = \frac{1}{\sin C} = \frac{\text{Hypotenuse}}{\text{Side opposite } \angle C}$
 $= \frac{AC}{AB}$
- ❖ $\sec C = \frac{1}{\cos C} = \frac{\text{Hypotenuse}}{\text{Side adjacent } \angle C}$
 $= \frac{AC}{BC}$
- ❖ $\cot C = \frac{1}{\tan C} = \frac{\text{Side adjacent to } \angle C}{\text{Side opposite to } \angle C}$
 $= \frac{BC}{AB}$

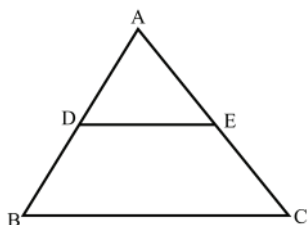
Thales Theorem (Basic Proportionality Theorem)

Statement: If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, then the other two sides are divided in the same ratio.

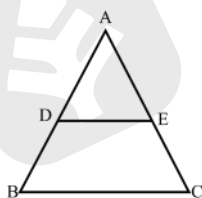
ie,

A triangle ABC in which a line parallel to side BC intersects other two sides AB and AC at D and E , respectively. Then

$$\frac{AD}{BD} = \frac{AE}{EC}.$$



Corollary: If in a $\triangle ABC$, a line $DE \parallel BC$ intersects AB at D and AC at E , then

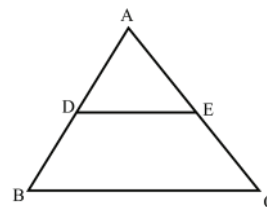


- | | |
|---------------------------------------|--------------------------------------|
| (i) $\frac{DB}{AD} = \frac{EC}{AE}$ | (ii) $\frac{AB}{AD} = \frac{AC}{AE}$ |
| (iii) $\frac{AD}{AB} = \frac{AE}{AC}$ | (iv) $\frac{AB}{DB} = \frac{AC}{EC}$ |
| (v) $\frac{DB}{AB} = \frac{EC}{AC}$ | |



Converse of Basic Proportionality Theorem

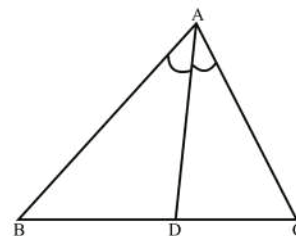
If a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.



ie., if in the figure, $\frac{AD}{DB} = \frac{AE}{EC}$ then, $BC \parallel DE$

Interior Angle Bisector Theorem

An angle bisector within a triangle divides the opposite side into two segments that are proportional to the other two sides of the triangle.

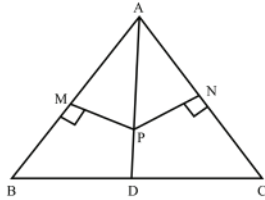


ie., if $\angle BAD = \angle CAD$ then $\frac{BD}{DC} = \frac{AB}{AC}$



Converse of Angle Bisector Theorem

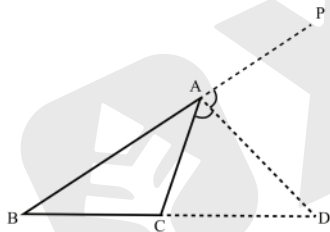
In a triangle, if the interior point is equidistant from the two sides of a triangle then that point lies on the angle bisector of the angle formed by the two line segments.



ie, if $PM = PN$ then AD is the angle bisector of $\angle BAC$

The External Angle Bisector Theorem

In a triangle, the external angle bisector of an angle divides the opposite side externally in the ratio of the other two sides of the triangle.



ie, if in $\triangle ABC$, AD is external angle bisector of $\angle BAC$ then

$$\frac{BD}{DC} = -\frac{AB}{AC}$$

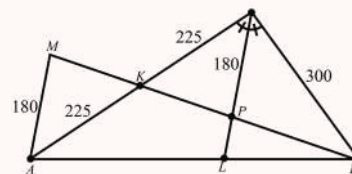


10. In a $\triangle ABC$, D and E are points on the sides AB and AC , respectively, such that $DE \parallel BC$.

If $AD = 4x - 3$, $AE = 8x - 7$, $BD = 3x - 1$ and $CE = 5x - 3$, find the value of x .



11. Triangle ABC has $AC = 450$ and $BC = 300$. Points K and L are located on $\overline{AC}$ and $\overline{AB}$ respectively so that $AK = CK$, and $\overline{CL}$ is the angle bisector of angle C . Let P be the point of intersection of $\overline{BK}$ and $\overline{CL}$, and let M be the point on line BK for which K is the midpoint of $\overline{PM}$. If $AM = 180$, find LP .



[AIME]



Area of a Triangle

Equations of Area of a Triangle

❖ $\Delta = \frac{1}{2} \cdot \text{base} \cdot \text{height}$

❖ Heron's Formula

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)},$$

where s = semi perimeter

❖ $\Delta = \frac{1}{2}ab \sin C$

❖ $\Delta = r \cdot s,$

where r = inradius.

❖ $\Delta = \frac{abc}{4R},$

where R = circumradius

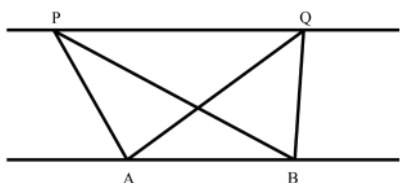
❖ $\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Where (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are vertices.

❖ $\Delta = \frac{\sqrt{3}}{4}a^2,$ for equilateral triangle

where a = side length.

Theorem: Two triangles on the same base (or equal bases) and between the same parallels are equal in area.

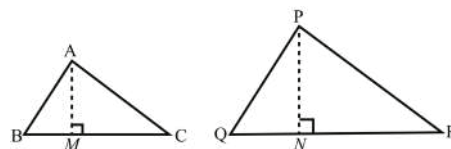


$$AB \parallel PQ \Rightarrow \text{ar}(\triangle ABP) = \text{ar}(\triangle ABQ)$$

Areas of Similar Triangles

Theorem: The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

Two triangles ABC and PQR , are such that $\triangle ABC \sim \triangle PQR$ [Shown in the figure]



Then:

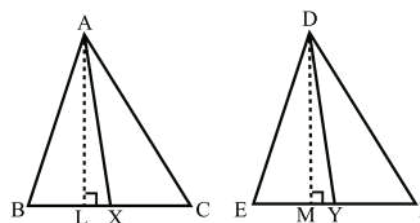
$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{RP}\right)^2.$$

Properties of Areas of Similar Triangles

❖ The areas of two similar triangles are proportional to the squares of their corresponding sides.

If $\triangle ABC \sim \triangle DEF$, then

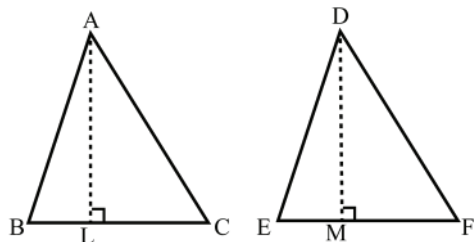
$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{AB^2}{DE^2} = \frac{BC^2}{EF^2} = \frac{AC^2}{DF^2}$$



- ❖ The areas of two similar triangles are proportional to the squares of their corresponding altitude.

If $\triangle ABC \sim \triangle DEF$, $AL \perp BC$ and $DM \perp EF$, then

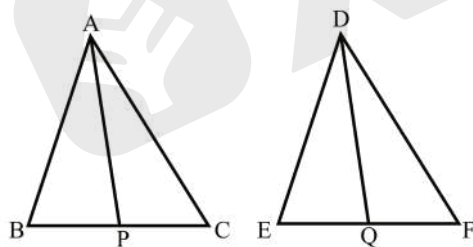
$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{AL^2}{DM^2}$$



- ❖ The areas of two similar triangles are proportional to the squares of their corresponding medians

If $\triangle ABC \sim \triangle DEF$, AP , DQ are their medians, then

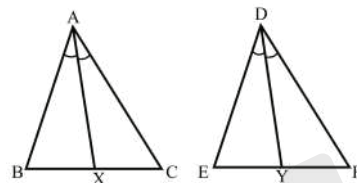
$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{AP^2}{DQ^2}$$



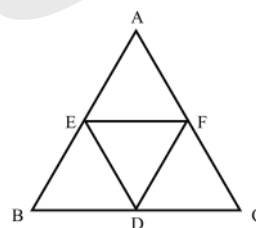
- ❖ The areas of two similar triangles are proportional to the squares of their corresponding angle bisector segments.

If $\triangle ABC \sim \triangle DEF$ and AX , DY are bisectors of $\angle A$ and $\angle D$, respectively, then

$$\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} = \frac{AX^2}{DY^2}$$



- ❖ If D , E and F are, respectively, the mid-points of sides BC , CA and AB of an equilateral triangle ABC then $\triangle DEF$ is also an equilateral triangle.



12. In triangle ABC , $\angle BAC = 90^\circ$, AD is its bisector. If $DE \perp AC$, prove that $DE \times (AB + AC) = AB \times AC$.





13. In a triangle ABC , let E be the midpoint of AC and F be the midpoint of AB . The medians BE and CF intersect at G . Let Y and Z be the midpoints of BE and CF respectively. If the area of triangle ABC is 480, find the area of triangle GYZ .

[IOQM 2023]



15. Let ABC be an acute-angled triangle and let H be its orthocentre. Let G_1, G_2 , and G_3 be the centroids of the triangles HBC, HCA , and HAB , respectively. If the area of triangle $G_1G_2G_3$ is 7 units, what is the area of triangle ABC ?

[PRMO 2018]



14. If 15 and 9 are the lengths of two medians of a triangle, what is the maximum possible area of the triangle to the nearest integer?

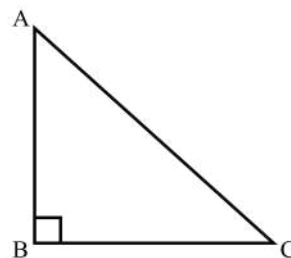
[PRMO 2019]



Pythagoras Theorem

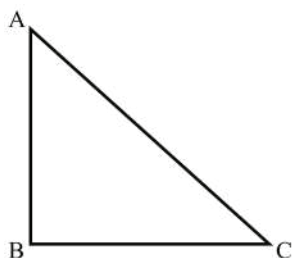
Statement: In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

In the figure: $AC^2 = AB^2 + BC^2$



Converse of Pythagoras Theorem

Theorem: In a triangle, if the square of one side is equal to the sum of the squares of the other two sides, then the angle opposite to the first side is a right angle.

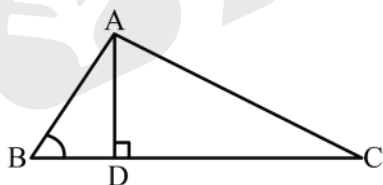


$$AC^2 = AB^2 + BC^2 \Rightarrow \angle B = 90^\circ$$

Some Results Deduced From Pythagoras Theorem

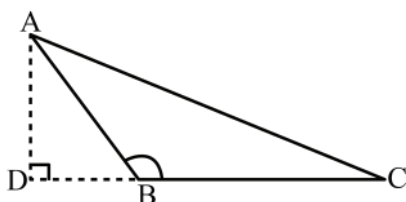
- ❖ In an acute angled $\triangle ABC$, AD is a perpendicular dropped on the opposite side of $\angle A$, then

$$AC^2 = AB^2 + BC^2 - 2BD \cdot BC \quad (\angle B < 90^\circ)$$



- ❖ In an obtuse angled $\triangle ABC$, AD is perpendicular dropped on BC produced to D , then

$$AC^2 = AB^2 + BC^2 + 2BD \times BC \quad (\angle B > 90^\circ)$$



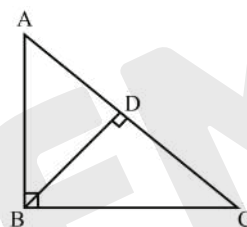
- ❖ In a right-angled $\triangle ABC$, $\angle B = 90^\circ$ and AC is the hypotenuse, the perpendicular BD is dropped on hypotenuse AC from vertex B , then

$$(i) \quad BD = \frac{AB \times BC}{AC}$$

$$(ii) \quad AD = \frac{AB^2}{AC}$$

$$(iii) \quad CD = \frac{BC^2}{AC}$$

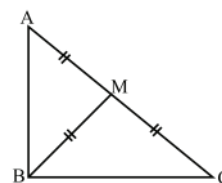
$$(iv) \quad \frac{1}{BD^2} = \frac{1}{AB^2} + \frac{1}{BC^2}$$



- ❖ In a right-angled triangle, the median to the hypotenuse

$$= \frac{1}{2} \times \text{hypotenuse},$$

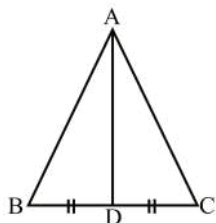
$$\text{i.e., } BM = \frac{AC}{2}$$



Apollonius Theorem

In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side.

$$\text{i.e., } AB^2 + AC^2 = 2(AD^2 + BD^2)$$



BD=CD, AD is the median



16. BL and CM are medians of $\triangle ABC$ right angled at A . Prove that

$$4(BL^2 + CM^2) = 5BC^2.$$



17. In a triangle ABC , the median from B to CA is perpendicular to the median from C to AB . If the median from A to BC is 30, determine $(BC^2 + CA^2 + AB^2) / 100$.

[PRE-RMO 2018]



Mid-Point Theorem

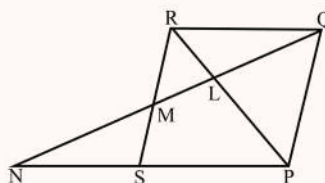
In a triangle, the line segment joining the mid-points of any two sides is parallel to the third side and is half of it.

Converse of the Mid-Point Theorem

The line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side.



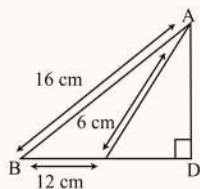
18. In the figure, $PQRS$ is a parallelogram with $PQ = 16$ cm and $QR = 10$ cm. L is a point on PR such that $RL : LP = 2 : 3$. QL produced meets RS at M and PS produced at N .



Find the lengths of PN and RM .



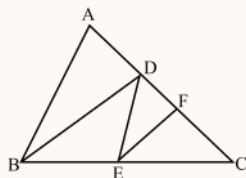
19. In the figure $\angle D = 90^\circ$, $AB = 16$ cm, $BC = 12$ cm, and $CA = 6$ cm, then CD is:



- (A) $\frac{13}{6}$ cm (B) $\frac{17}{6}$ cm
(C) $\frac{19}{6}$ cm (D) $\frac{18}{5}$ cm



20. In the given figure, $\triangle ABC$ has points D and F on $\overline{AC}$ and point E on $\overline{BC}$ such that $\overline{DE} \parallel \overline{AB}$ and $\overline{EF} \parallel \overline{BD}$. If $\overline{CF} = 4$ cm and $\overline{AC} = 9$ cm, what is the length of $\overline{DC}$?



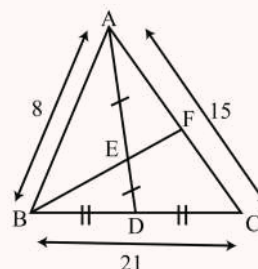
- (A) 7 cm (B) 6 cm
(C) 5 cm (D) 4 cm



21. In $\triangle ABC$, AD is the median through A and E is the mid-point of AD . BE produced meets AC at F . Prove that $AF = \frac{1}{3}AC$.



22. In $\triangle ABC$, AD is median and E is the mid-point of AD . If BE is extended, it meets AC at F . $AB = 8$ cm, $BC = 21$ cm, and $AC = 15$ cm, then $AF = ?$



- (A) 7cm (B) 3cm
(C) 12cm (D) 5cm



23. In a triangle ABC , the median AD (with D on BC) and the angle bisector BE (with E on AC) are perpendicular to each other. If $AD = 7$ and $BE = 9$, find the integer nearest to the area of triangle ABC .

[PRMO 2019]

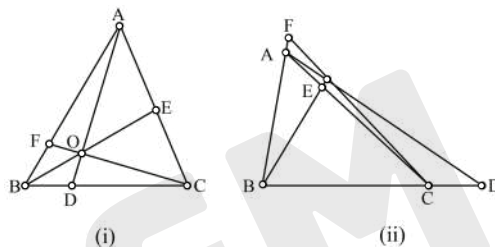


Ceva's Theorem

Theorem (Ceva)

If points D, E and F are taken on the sides BC, CA and AB of $\triangle ABC$ so that the lines AD, BE and CF are concurrent at a point O , then

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1 \quad \dots(i)$$



Converse of Ceva's Theorem

If three points D, E and F taken on the sides BC, CA and AB of $\triangle ABC$ are such that equation (i) holds, then AD, BE and CF are concurrent.

Trigonometric form of Ceva's Theorem

Let points D, E and F be taken on the sides BC, CA and AB of $\triangle ABC$. Then the lines AD, BE and CF are concurrent if and only if

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1$$

Corollary 1. The medians of a triangle are concurrent.

Corollary 2. The altitudes of a triangle are concurrent.

Corollary 3. The internal bisectors of the angles of a triangle are concurrent.

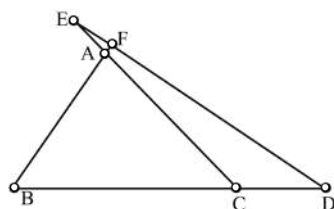
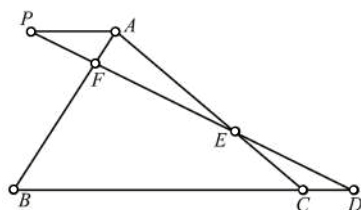
Corollary 4. The external bisectors of any two angles of a triangle and the internal bisector of the third angle are concurrent.

Corollary 5. If the incircle of $\triangle ABC$ touches BC, CA, AB at X, Y, Z , respectively, then AX, BY, CZ are concurrent (the point of concurrence is called the Gergonne point of $\triangle ABC$)

Menelaus' Theorem

If a transversal cut the sides BC, CA, AB (suitably extended) of $\triangle ABC$ in point D, E, F , respectively, then

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1 \quad \dots(ii)$$



Converse of Menelaus' Theorem

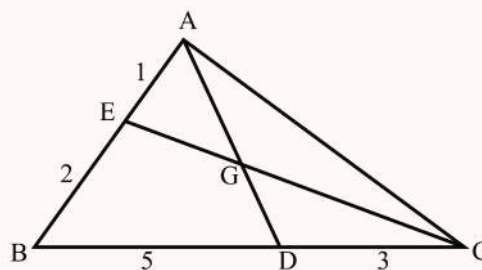
If points D, E, F are taken on the sides of $\triangle ABC$ such that equation (ii) holds, then D, E, F are collinear.



24. On the sides BC, CA and AB of $\triangle ABC$ point D, E, F are taken in such a way that $\frac{DB}{DC} = \frac{CE}{EA} = \frac{AF}{FB} = 2$. If the area of the triangle determined by the line AD, BE, CF is $\frac{1}{K}$ times of area $\triangle ABC$ find K .



25. In the figure $AE : EB = 1 : 2$ and $BD : DC = 5 : 3$. Find $EG : GC$.





Mass Point Geometry

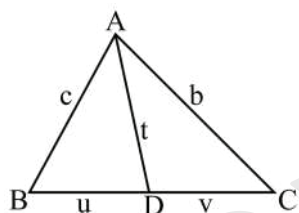


$$\therefore m_1 l_1 = m_2 l_2$$

$$\text{i.e., } \frac{m_1}{m_2} = \frac{l_2}{l_1}$$

The seesaw will balance if the product of the mass and its distance to the fulcrum is the same for each mass

Stewart's Theorem



- (i) If u, v are the actual length of BD, DC respectively then

$$t^2 = \frac{ub^2 + vc^2}{u + v} - uv$$

- (ii) If $BD : DC = u : v$ then,

$$ub^2 + vc^2 = (u + v)t^2 + u \cdot v^2 + v \cdot u^2$$

Note:

Apollonius Theorem is special case of Stewart's Theorem.



26. ABC is a triangle, D and E are interior points of sides AB and BC , respectively, are such

that $\frac{AD}{DB} = \frac{1}{3}, \frac{DE}{EB} = \frac{1}{3}$, if AE and CD

intersect at F . Find $\frac{CF}{FD}$.

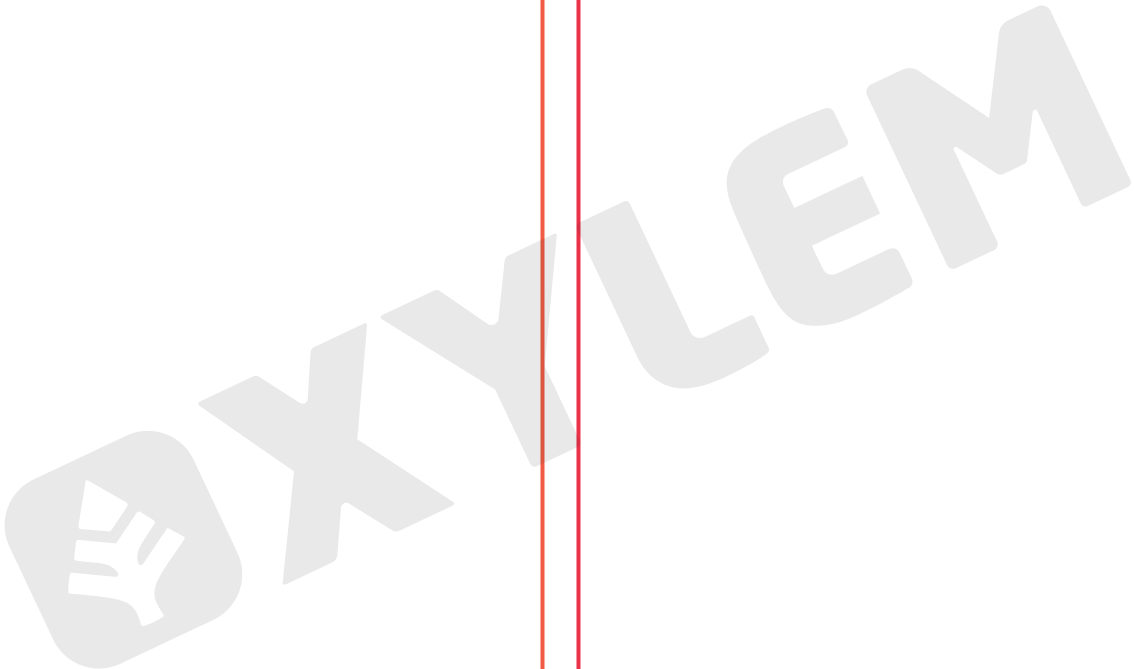


ANSWER KEY

(1) -	(7) -	(14) 90	(20) (B)
(2) 48°	(8) -	(15) 63	(21) -
(3) 40°	(9) 69	(16) -	(22) (D)
(4) 260°	(10) 1	(17) 24	(23) 47
(5) $\angle TQR = 20^\circ$ $\angle QTR = 130^\circ$	(11) 72	(18) $PN = 15$ $RN = 10.67$	(24) 7
(6) (ii)	(12) -	(19) (C)	(25) 5 : 9
	(13) 10		(26) 12

Space for rough work

Space for rough work



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